Session 7: New tools, your own question

Question: What question will you answer in your project?

Concepts

Concept	In daily life	In Excel	The limit
SQL	Asking a database a question	A filter and PivotTable in words	Same ideas as pandas, different spelling
SELECT / FROM	Choose the columns, name the table	Choose columns	Text goes in single quotes
WHERE	Keep only the rows you want	Filter	Use = for equal, not ==
GROUP BY, ORDER BY, LIMIT	Pile, sort, keep the top	PivotTable, sort	ORDER BY total DESC puts the biggest first
Power BI	A dashboard tool	Charts and cards	It shows data. Python prepares and checks it

Code patterns

```
SELECT country, SUM(revenue) AS total_revenue
FROM sales
GROUP BY country
ORDER BY total_revenue DESC
LIMIT 5

SELECT SUM(revenue) FROM sales WHERE country = 'Germany'

totals = sales_clean.groupby("country")["revenue"].sum()   # the same question in pandas
```

Three common errors

Error	What to do
no such table	The table name must be the same as in to_sql("sales", ...).
no such column	Check the column name. It must be one of the seven we put in the table.
Syntax error near ...	Check for a missing comma or a quote. Text needs single quotes.

Exercises and expected answers

Exercise	Expected
Predict: Read the query first. What will it return: how many rows, which columns? Then run it.	Guess first, then run it to check
Fill in: Total revenue of Germany. In SQL, WHERE is the filter and text goes in single quotes.	37365.63
Fill in: The 3 rows with the biggest revenue.	the first row is invoice 540815 with 6539.4
Fix it: This query forgot which table to ask. Fix it.	1876534.04
Write it: How many different orders did Germany place? Use COUNT(DISTINCT invoice).	79, the same as Session 4
Write it: How many countries are in the data? Use COUNT(DISTINCT country). Then check it with pandas nunique().	31 both times
Case: Menu option 2 starts here. The top 5 products of the Netherlands by revenue, in SQL.	ROUND SNACK BOXES SET OF4 WOODLAND on top with 1958.4
Extra: Revenue per country for countries with more than 30000 pounds only. In SQL, a filter on a group uses HAVING.	5 countries
Top 5 countries by query	United Kingdom, Netherlands, EIRE, Germany, France
Germany revenue	37365.63
Top 10 revenue rows (homework)	The first row has revenue 6539.4
Your project plan	One question, columns, steps, a chart, a decision

Project plan

Part	Your answer
My question	
Columns I need	
Steps	
The chart I expect	
The decision it helps	